# Supplementary material

Parallelizing Quantized LLM Inference
Under a Fixed Numerical Reference

Chengzhan Li
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This supplement supplies implementation coverage, complete comparisons and reproducible source identities for the main paper. Numerical observations refer to their recorded implementations and inputs. The reconstruction accompanying this draft uses CPU analysis of public records; it adds no native qualification or model timing.

## A Numerical contract and implementation coverage

### A.1 What is bound and what is checked

The reference binds the producer as well as the consumer. In the CUDA deployment, the original prompt producer uses physical microbatch one and requests its original final output. Candidate prefill can regroup known rows when the resulting requested vector and active cache preserve that producer. The Vulkan companion retains its original physical microbatch 128 and outer prompt chunks. Equal emitted-token endpoints alone do not make states produced by different schedules interchangeable.

The dense append check in `phase_append_requirements` requires one stream, no sliding window or shifted cells, an occupied prefix whose end and physical head equal the append position, and sufficient capacity. Existing cells must have their own logical positions and include sequence zero. The native padding is at least 256 entries. The executor additionally binds F16 K/V and a context owned by its single-sequence frontend. The sequence-membership check is not an exclusivity test for externally mutated, shared caches.

The phase executor accepts all-row or last-row output requests. The original released frontend reserves four output rows; its all-output gate rejects wider requests before a native call, although last-only prompt calls can be wider. A separate eight-row-reserve profile must be qualified with its own consumer. The planner's more general requested-row representation does not turn the original frontend into an arbitrary sparse-output API.

Local attention requires causal, non-Flash execution, consecutive positions, one sequence, a complete prefix-extent vector, singleton K/V streams, and no attention bias, sinks or MLA path. Its descriptor includes Q/K/V types, head dimensions and counts, and V transposition. Equal-descriptor runs use at most four query columns in the admitted layout, with K, V and mask views restricted to the required extent. A noncontiguous mask is materialized before use. Several architecture and layout requirements are assertions rather than recoverable admission statuses.

### A.2 Projection checks and reuse

The Q8 packed branch accepts F32 activations with Q8_0 weights and query width 5–32. Actual multi-group packing additionally requires contiguous activations, singleton higher weight dimensions and at least eight queries. It maps complete groups to $[K, 4, \lfloor q/4 \rfloor]$, reshapes the outputs, and handles a remaining tail with an ordinary operation of at most four columns. Widths five through seven therefore enter this branch without constructing multiple packed groups.

The marked backend-loop4 control checks contiguous Q8_0/F32/F32 tensors, dimensions, offset bounds and absence of fused arithmetic or indexed matrix multiplication. Its CUDA dispatcher requires actual buffers on the selected device, NVIDIA warp size 32, safe padding, and MMVQ selection for every original one-to-four-column tile. These checks precede that operator's quantizer and kernel, but occur inside a decode call; earlier graph operations can already have executed.

Packed projections have no loop4 marker. Their ordinary dispatcher can select other kernels, including a cuBLAS branch for unsuitable padded views. Falling back to an ordinary graph path is not, by itself, a reference-equivalence guarantee. The mixed Q4_K/Q6_K experiment has its own one-column control and is outside these Q8 marker checks.

The active graph-reuse checks compare the policy, full prefix-extent vector, head-placement key, and normal graph topology and output count. Requested row indices are refreshed as graph inputs. A structural identity helper exists in the planner header but is not the live reuse key. Accordingly, neither a helper's presence nor a cache-type tag supplies a general run-time certificate for every numerical choice.

### A.3 Output lifetime, observation and failures

After synchronization, each requested vector is copied to storage indexed by its logical row. A later group can then reuse native scratch buffers without destroying an earlier accepted frontier. The implementation checks that a requested output exists and that the physical append endpoint is correct; it does not compare reference KV bytes online. Greedy sampling checks all vector entries for finiteness and retains the first vocabulary index on ties.

The original capture observer selects attention and specially named nodes; its filter does not include the packed projection names. Capture is disabled during clean timing. The original state-capture call records initialization and the first commit. Each other profile must be assessed from its own capture implementation and qualification records. A native sequence export also requires extraction of active K/V before it represents the reference relation in the main paper.

The target decode loop poisons its owner on exceptions from native groups and postchecks. Trim, reset, draft, sample and diagnostic failures lie outside that catch; the command-line entry exits on failure. Assertions terminate directly. The resulting process boundary prevents another request in that worker, but does not establish recovery for a library caller that catches an exception and continues.

### A.4 Diagnostics that distinguish consumer contracts

Two CPU diagnostics demonstrate why output and state requirements must be declared together. They use a synthetic two-layer F32 model, F16 KV, a vocabulary of 128, one CPU thread and a context capacity of 1024. They are mechanism diagnostics, not GPU qualification or performance measurements [1].

First, a two-row call and two successive one-row calls start from identical state at history length 248. Although both operation argmax values and the next forced-continuation argmax agree, 242 of 256 output float bit patterns differ. Their final serialized active-sequence states differ in 43 bytes, and the continuation differs in 127 of 128 float bit patterns. The diagnostic does not localize the cause. It demonstrates that token agreement can coexist with different full vectors, state and subsequent computation.

Second, an eight-row operation is compared with a predefined four-plus-four reference at history lengths 248 and 252. All eight shared outputs, active state, geometry and one forced continuation agree. A restricted downstream consumer instead requires the first four vectors, retains five materialized rows, and then consumes fixed token 37. This comparison uses a four-plus-one reference and trims the wider execution to the same five-row frontier. Extra vectors are captured for diagnosis, but the fifth vector is excluded from the downstream contract before execution. It differs in 117 and 128 entries at the two histories, respectively, despite unchanged argmax; the required four outputs, retained state and forced consumer agree. Thus this restricted consumer passes while one requiring the fifth vector does not. These are explicit diagnostic schedules, not an autonomous scalar deployment reference. They establish neither natural speculative acceptance behavior nor a general qualification of an eight-output production verifier.

## B Workloads, numerical coverage and statistical reconstruction

The CUDA references bind a whole-prompt caller with physical microbatch one and last-only prompt output; the Vulkan companion retains its original microbatch 128 and outer prompt calls. The main configurations use one dense nonwrapping sequence, F16 K/V, greedy generation and non-Flash causal attention. Known-query work uses up to 32 positions. A different model, device, producer or numerical module defines a different reference.

The historical Linux Qwen/Smol reports describe comparisons of complete requested vectors and active K/V by layer and logical position. Their geometry cases cover widths 1, 4, 8 and 32, inside and crossing prefixes, tails and the tested output masks. Natural cases exercise early and later rejection, trim, pending correction and bonus tokens, native end-of-generation, caps and reset. The Llama deployment has a separately established reference. These historical observations are retained as reported qualification, rather than an independent replay of the original native records. The 32B strong-control record directly identifies full 151,936-entry output and 64-layer active-state comparisons at its tested boundaries.

The wide-verification study initially freezes source, order and analysis before 1,024 clean timings and 128 primers. A mask-reader correction is validated offline and the completed first worker is retained without rerunning the GPU; its gap remains in the worker accounting. A later joint boundary/rollback qualification on an exposed natural root adds 1,936 full-vector and 296 active-target-KV comparisons across all eight arms. These later observations extend the checked transitions, not the diagnostic coverage of the earlier clean runs.

APPS and MATH provide the natural Qwen task identities [2, 3]. Qwen 8B and 14B share 16 roots and four sessions. Llama uses 12 roots selected after excluding previous project exposure. Each six-arm panel has

two separate primer requests per worker; its 48 primers are not inference units. Four-arm factorials use four Williams rows, with 256 formal requests and 32 primers per model. The original 8B two-D4 order is fixed; its compound contrast therefore retains an order confound that the later balanced factorial does not. Model hashes, prompt IDs and orders are linked in the source index.

The 32B strong comparison uses 12 roots, six coding and six mathematics, four rounds and three arms. Its 144 measured requests each follow a matched warmup. Root-level paired log ratios are averaged equally; the bootstrap resamples the two six-root strata separately. Its primary C/B interval uses 95% coverage and secondary G comparisons 97.5%. Q8 application and factorial intervals use 99% coverage for their five-comparison families. Repeated sessions travel with the root under resampling. These intervals condition on the observed sessions.

The natural multi-turn pilot uses two OpenAssistant development conversations [4], two rounds and eight measured episodes. Its canonical transition does not initially retain the currently generated body as reusable KV; an earlier canonical prefix can be reused on later turns. This protocol creates less local-cut opportunity than assuming that every generated body persists. Its pooled ratio is descriptive. The dataset revision's card and license identify Apache-2.0; the manuscript package redistributes timing records rather than conversation text.

**Reconstruction levels.** The supplied 8B/14B/Llama service and factorial files contain 1,568 measured timing rows. The 32B strong file includes 144 warm and 144 measured rows; the multi-turn file has eight measured episodes. The CPU routines preserve original root order, bootstrap seeds and estimands. Separate NumPy and standard-library implementations agree to $2 \times 10^{-12}$ in the reconstructed statistics. This tolerance concerns arithmetic in the statistical analysis, not numerical compatibility of inference. Other legacy panels supply the public numeric projections or root aggregates identified in their manifests; availability of those projections does not imply availability of all native receipts.

# C Complete application and component comparisons

Table C.1: Mean resident seconds per request for all six A5000 application arms. Each Qwen arm contains 64 measured requests, and each Llama arm 48. W means SIMPLE whole-graph cuts; L means LOCAL; L4 is backend-loop4 and P is packing. M loads target resources alone. D4 uses draft prefill microbatch four with $k = 3$.

| Model | W-L4-M | W-P-M | L-L4-M | L-P-M | W-L4-D4 | L-P-D4 |
|---|---|---|---|---|---|---|
| Qwen3-8B | 2.727386 | 2.689778 | 2.728748 | 2.691337 | 2.272642 | 2.226927 |
| Qwen3-14B | 4.629760 | 4.602216 | 4.629086 | 4.602757 | 3.464285 | 3.426847 |
| Llama-8B | 2.887876 | 2.851631 | 2.887360 | 2.851517 | 2.563052 | 2.522963 |

The Windows practical comparison gives 29.11% less resident time and 26.00% less worker time than the fastest fixed M. It includes the complete speculative application. Its phase account saves 96.889 s of target work, adds 31.931 s of draft work and 0.962 s of other costs, for 63.996 s net resident savings. Device-memory samples show an additional 0.705 GiB peak for the draft-containing configuration; per-process memory peaks and energy were not measured.

The B570 resident pilot instead favors ordinary generation. Its four exposed roots and two blocks give mean times of 1.1503 s for M and 1.8235 s for the fastest speculative arm. A separate mechanism record has 578 local and 582 whole target graphs for 1,156 target rows, compared with 992 scalar M graphs/rows. A distinct synchronized two-root cost diagnostic saves 0.466681 s in target work but adds 1.923330 s in draft work and 0.210582 s elsewhere. The different windows are not combined.

Grouping improves this earlier panel by 5.62% with whole cuts and 5.61% with local cuts relative to its graph-loop control. Those effects include shared-quantization advantages that the main paper's stronger B control already obtains. Grouped D4 is 14.00% slower than the fastest ordinary grouped arm. The strong-control operator follow-up separately uses three paired blocks in four exposed tensor cells, with one warmup and one measured operation per fresh worker. Table C.6 retains all cells, including mixed directions for Q6_K/$q = 4$.

Table C.2: All five fixed application comparisons in each A5000 model. The numerator is L-P-D4 throughout. Each interval is a 99% paired-root interval from that model's five-comparison family. Ratios use total resident time.

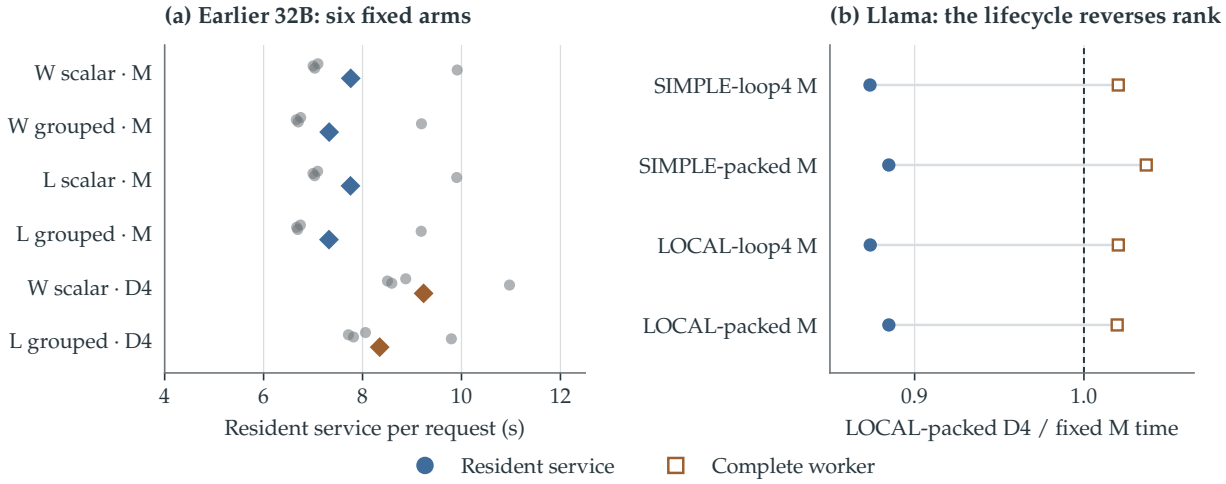
| Target | Fixed comparator | Time ratio | 99% lower | 99% upper |
|---|---|---|---|---|
| Qwen3-8B | W_L4_M | 0.816506 | 0.747568 | 0.872304 |
| | W_P_M | 0.827922 | 0.753870 | 0.888346 |
| | L_L4_M | 0.816098 | 0.747171 | 0.872569 |
| | L_P_M | 0.827443 | 0.753938 | 0.887542 |
| | W_L4_D4 | 0.979884 | 0.975589 | 0.985559 |
| Qwen3-14B | W_L4_M | 0.740178 | 0.648295 | 0.811920 |
| | W_P_M | 0.744608 | 0.651094 | 0.817914 |
| | L_L4_M | 0.740286 | 0.648446 | 0.812100 |
| | L_P_M | 0.744521 | 0.651113 | 0.817854 |
| | W_L4_D4 | 0.989193 | 0.987176 | 0.991690 |
| Llama-8B | W_L4_M | 0.873640 | 0.783602 | 0.941788 |
| | W_P_M | 0.884744 | 0.790475 | 0.956649 |
| | L_L4_M | 0.873796 | 0.783749 | 0.941931 |
| | L_P_M | 0.884779 | 0.790486 | 0.956653 |
| | W_L4_D4 | 0.984359 | 0.981680 | 0.988452 |

Table C.3: All four factorial effects and the interaction in each Qwen model. Direct effects are ratios of total service; the interaction is an equal-root geometric ratio of ratios. Intervals are 99% paired-root percentiles in the five-comparison family.

| Model | Comparison | Ratio | Interval |
|---|---|---|---|
| 8B | L_L4_D4/W_L4_D4 | 0.997913 | [0.995370, 0.999576] |
| 8B | L_P_D4/W_P_D4 | 0.998717 | [0.995953, 1.000570] |
| 8B | W_P_D4/W_L4_D4 | 0.983648 | [0.980236, 0.988654] |
| 8B | L_P_D4/L_L4_D4 | 0.984440 | [0.981228, 0.989391] |
| 8B | Interaction | 1.000696 | [1.000069, 1.001393] |
| 14B | L_L4_D4/W_L4_D4 | 0.996957 | [0.994726, 0.999092] |
| 14B | L_P_D4/W_P_D4 | 0.998881 | [0.997093, 1.000227] |
| 14B | W_P_D4/W_L4_D4 | 0.990130 | [0.988351, 0.992804] |
| 14B | L_P_D4/L_L4_D4 | 0.992041 | [0.990754, 0.993985] |
| 14B | Interaction | 1.001703 | [1.000293, 1.003338] |

Table C.4: Lifecycle costs for selected fixed controls. Resident, burst and worker clocks overlap and must not be added. A dash is an omitted field, not a zero. Workers include loading, integrity checks and primers; OS file caches were not cleared, so these are not cold-start latency claims.

| Deployment | Comparison | Resident (s) | Worker (s) | Interpretation |
|---|---|---|---|---|
| A5000 Qwen3-8B | W-P-M / L-P-D4 | 172.146 / 142.523 | 200.739 / 168.973 | Published worker totals are rounded to milliseconds. |
| A5000 Qwen3-14B | W-P-M / L-P-D4 | 294.542 / 219.318 | 662.546 / 594.559 | Measured-request bursts are 294.632 / 219.405 s. |
| A5000 Llama-8B | L-P-M / L-P-D4 | 136.873 / 121.102 | 361.717 / 368.822 | The fastest M worker is W-P-M at 355.730 s, a different arm from fastest resident M. |
| Windows Qwen3-8B | W-P-M / L-P-D4 | 219.810 / 155.814 | 260.240 / 192.568 | Separate deployment; input identities later reused on A5000. |




Four exposed 32B roots. Worker totals include process/load/primer work; source values are rounded.

Figure C.1: Earlier mixed-Q4 application results and the lifecycle reversal. Panel (a): the four-root 32B panel uses a graph scalar-loop control and is distinct from the later 12-root strong-backend study; all six paths include slower speculative generation. Panel (b): Llama-8B resident and worker costs are shown for the four fixed M controls and the candidate; overlapping clocks are not stacked. Source values are public numeric projections, identified in the manifest.

Table C.5: All six earlier Qwen3-32B mixed-Q4 arms: four exposed mathematics roots, four rounds, 16 measured requests per arm and 16 identical-root warmups. REF1L is the graph scalar-column loop and REF1G is independent grouping. The subsequent strong B control is absent from this earlier panel.

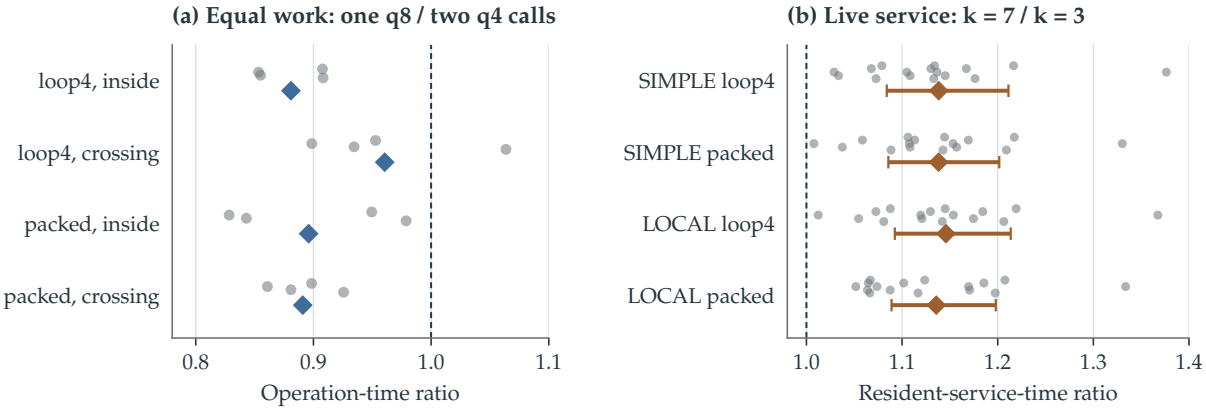
| Arm | Resident sum (s) | Mean (s) |
|---|---|---|
| W_REF1L_M | 124.182883 | 7.761430 |
| W_REF1G_M | 117.208239 | 7.325515 |
| L_REF1L_M | 124.100017 | 7.756251 |
| L_REF1G_M | 117.143582 | 7.321474 |
| W_REF1L_D4 | 147.748374 | 9.234273 |
| L_REF1G_D4 | 133.543739 | 8.346484 |

Table C.6: Strong scalar-loop operator follow-up. B/G and C/B are geometric means of three paired elapsed-time ratios; ranges show observed paired minima and maxima, not confidence intervals. Equal-cell geometric ratios are 0.6390 and 0.8528. These are operator observations, distinct from the later complete service result.

| Cell | B/G | C/B | C/B range |
|---|---|---|---|
| Q4_K/q4 | 0.737011 | 0.969303 | [0.957626, 0.988546] |
| Q4_K/q32 | 0.473109 | 0.692862 | [0.680304, 0.702919] |
| Q6_K/q4 | 0.820466 | 0.990319 | [0.962415, 1.010187] |
| Q6_K/q32 | 0.582934 | 0.795353 | [0.747288, 0.870814] |

# D Wider execution and workload cost

The Windows equal-work panel uses one APPS and three GSM8K origins [5], fixed initial state, the same eight known tokens and all eight requested outputs. Its 128 measured operations and 32 primers address operation efficiency. The separate live panel uses 16 roots, prompt length at most 192, cap 96, and eight balanced order rows. Both $k = 3$ and $k = 7$ reserve eight outputs; 1,024 measured requests and 128 primers cover eight arms. The four primary width comparisons use 98.75% paired-root intervals; secondary fixed-$k = 7$ comparisons form a separate family.




**(c) Live work changes even when the emitted output count is fixed**

| Verification cycles | Proposals | Verified rows | Emitted tokens |
|---|---|---|---|
| 3,504 → 2,296 | 10,512 → 16,072 | 14,016 → 18,368 | 11,472 → 11,472 |

Figure D.1: Physical execution width and live proposal width. Equal-work $q = 8$ versus two $q = 4$ operations is a fixed-input comparison; $k = 7$ versus $k = 3$ resident generation changes the work. The live summaries use ratios of totals and 98.75% paired-root intervals over 16 roots, with repeats clustered by root. Work counts refer to the same emitted-token total. These panels intentionally have different measurement boundaries.

Each live arm emits 11,472 tokens. For $k = 3/k = 7$, proposed tokens are 10,512/16,072, accepted proposals 7,928/9,152, first inspected mismatches 1,136/1,256, and unconsumed suffix tokens 1,448/5,664. Native cap/end-of-generation counts are 104/24 in each arm. Suffix waste includes post-rejection and terminal-boundary work; the aggregate counts do not separate these causes.

The A5000 four-root pilot has 64 live measurements and 32 primers. Its LOCAL-packed $k = 7/k = 3$ ratio is 1.140572. At fixed LOCAL and $k = 7$, packing/loop4 is 0.981741. A separate proposal-tape replay excludes draft work and has different target-row counts between widths, so it is not a complete-generation comparison.

Table D.1: A5000 live-width pilot. Entries are resident totals over four exposed roots and two rounds; no population interval is inferred.

| Cut / projection | $k = 3$ (s) | $k = 7$ (s) |
|---|---|---|
| W_L4 | 12.038851 | 13.824167 |
| W_P | 11.897263 | 13.598468 |
| L_L4 | 11.996725 | 13.800486 |
| L_P | 11.878694 | 13.548505 |

An initial A5000 equal-work comparison had unmatched first-use histories. The subsequent comparison uses 16 fresh processes, each with four warm and four measured operations, and balanced ABBA/BAAB preparation histories. It holds one exposed input, initial state and eight outputs fixed at two prefix coordinates. The complete 128-operation account gives the four process-level summaries below; mixed directions remain visible.

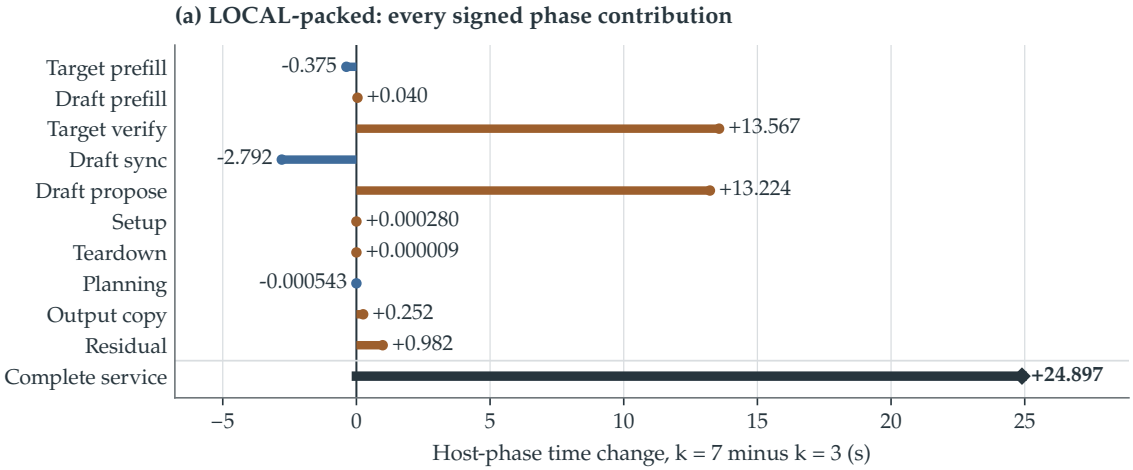



Figure D.2: Where the extra live cost arises. Disjoint phase differences for LOCAL-packed are $k = 7$ minus $k = 3$ over 128 requests. Negative components save time; the net increase is 24.897 s. Phase timers include their specified host work, synchronization and GPU execution; the residual closes the recorded service sum. These are complete-arm phase sums, not new native measurements.

Table D.2: Matched-history A5000 equal-work comparison. Each cell has four fresh processes on one input; q8/q4-pair geometric means and observed process ranges are descriptive.

| Projection | Prefix | Geometric ratio | Process range |
|---|---|---|---|
| loop4 | 248 | 0.990445 | [0.941449, 1.042653] |
| loop4 | 252 | 0.978065 | [0.952121, 1.023051] |
| packed | 248 | 0.936989 | [0.891884, 1.001187] |
| packed | 252 | 0.947566 | [0.925460, 0.977713] |

# E Boundary counterexamples and numerical migration

The final B570 append panel has 12 fixed cells, three paths and three paired repeats, with 108 measured operations and 108 corresponding warmups. At $q = 32$ it retains the registered last-only output request. Its timer begins after valid state preparation and includes phase binding, native decode, synchronization and copying the requested vector. Caller batch preparation and state import are outside it. Earlier geometries remain relevant because a $16 + 16$ crossing can preserve different surrounding work from a $1 + 31$ crossing.

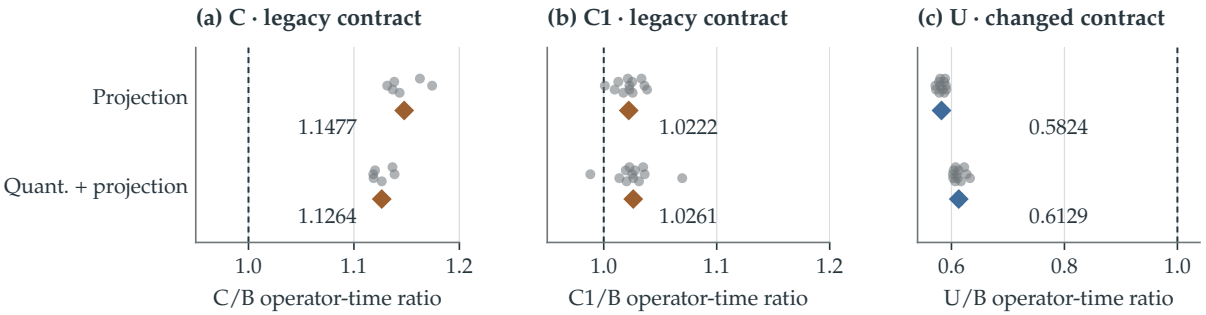
**Historical fixed-work geometry: LOCAL / SIMPLE time**

| | q = 2 | q = 4 | q = 8 | q = 16 | q = 32 |
|---|---|---|---|---|---|
| Qwen · C1024 dual resident | **0.822** inside 0.991 | **0.761** inside 0.999 | **0.838** inside 0.996 | **1.047** inside 1.000 | **1.055** inside 1.014 |
| Qwen · C4096 target only | **0.885** inside 1.000 | **0.882** inside 1.012 | **0.929** inside 0.990 | **1.010** inside 0.997 | **1.015** inside 0.999 |
| Smol · C4096 dual resident | **0.810** inside 1.011 | Not measured | **0.791** inside 1.016 | **0.923** inside 0.982 | **1.089** inside 1.001 |

All measured and unmeasured cells retained. These splits differ from the 1 + (q − 1) panel.

Figure E.1: Earlier local-cut geometries, including reversals. Every recorded cell and its inside control is retained; a missing cell is shown as unmeasured, not zero. These historical fixed-cell timings use their original measurement boundaries. They establish sensitivity to split geometry, not a population estimate or an isolated capacity effect.

The modern RTX comparison extracts a $4096 \times 4096$ Q8 projection with $q = 8$ from one captured activation. U uses the upstream quantizer and few-row projection [6]; B retains the old reference. Ordinary launches with CUDA graphs disabled and the recorded programmatic-launch setting are timed using GPU events. Allocation, copies and checks are excluded. The full quantization-plus-projection and projection-only boundaries have 12 paired repeats each, totaling 48 samples. Ratios are reconstructed from event duration divided by batch repetitions, avoiding rounding in the printed per-operation column.




One captured input; each panel is a separate experiment. Dots are paired blocks, not inputs.

U differs in 32,766/32,768 outputs. The complete modern backend was not timed.

Figure E.2: Extracted RTX operator alternatives under distinct numerical contracts. Legacy-compatible C and C1, and modern upstream U, come from separate sessions and are not directly paired with one another. Points retain all paired repeat ratios for each boundary. U/B complete-operator time is 0.612924 with a 95% paired-repeat interval [0.607906, 0.618665]; projection-only time is 0.582391. U does not preserve B's quantization or output bytes. The earlier old-oracle qualification failure and the unrun complete default backend have no timing point.

U's older-contract qualification fails before timing; the subsequent U-contract study compares a valid modern numerical alternative. Static API checks for a complete current default backend pass without loading

a model, but full-backend numerical qualification and service timing are unrun. These statuses cannot be promoted to measured service performance. The faster U operator and slower same-contract candidates are retained together so that changing the reference remains a visible engineering option.

# F Fixed source index and access scope

The source package contains `data/`, `scripts/`, `figures/` and `generated/`. The figure manifest records every included input's repository URL, commit and SHA-256; the figure metrics record the estimand and plotted values. `build_figures.py` produces editable SVG, vector PDF and preview PNG files. `build_tables.py` creates editable table rows. The CPU reconstruction scripts retain frozen seeds and reject incomplete panels. These scripts operate on public records and do not load a model or request a GPU.

Table F.1 supplies direct frozen entry points. Short revision labels are hyperlinks to full commit URLs. The machine-readable manifests preserve complete identifiers and per-file hashes. An experiment revision, a source snapshot and a reconstruction revision serve different purposes and are not interchangeable.

Table F.1: Frozen scientific sources. Public timing records support the stated statistical reconstructions; a code or result link does not promise access to every underlying native tensor.

| Evidence | Fixed revision | Direct entry and role |
|---|---|---|
| Physical-range intervention | `701079f2` | Fixed one-query AV crop/pad replay; complete operator output comparisons, with native input identity reported. Raw tensor archives were not replayed in this reconstruction. |
| Native grouping planner | `ad16a13d` | Dense-cache requirements, physical groups and structural helper. The live graph-reuse key is separately implemented in native code. |
| Original execution sources | Source snapshot | Published source snapshot; SHA-256 begins `57cbfbfe43ae1c4b`. It identifies frontend ownership, native local attention and packed/loop4 implementations. Full hash and member identities are in the method source map. |
| 32B strong-control code | `ab695a76` | Graph and MMVQ source differences for G, B and C; frozen scalar arithmetic and shared-quantization control. |
| 32B service result | `ab695a76` | Qualification scope, reconstructed timing rows, per-root ratios, statistical report and independent reconstruction. Main strong-baseline figure and 1.51% claim. |
| Public timing reconstruction | `10fc03ed` | Complete public Q8 timing rows, plans, service projections and CPU recomputation scripts. Source of the included `data/closeout/` files. |
| 8B service | `592abdfe` | Original request table, plan and six-arm result. The two speculative arms retain their fixed relative order. |
| 8B factorial | `55f501e9` | Four-arm Williams design, requests, component effects and interaction. |
| Llama confirmation | `ca2c769c` | Twelve selected roots and six-arm result; `evidence/inputs-01/MODELS.json` identifies target and draft bytes. |
| 14B service | `dbca51e5` | Six-arm service and worker costs on the shared Qwen roots. |
| 14B factorial | `ca6caee2` | Separately measured four-arm attribution; not pooled with the service panel. |
| Natural multi-turn pilot | `92c47c1a` | All eight timed episodes; two development conversations, two rounds and two arms. |
| Proposed phase analysis | `750346a1` | Source-derived 89-row example and local-cost decision; no new service timing. |
| Controlled B570 appends | `62f2e334` | Twelve geometry cells with paired inside controls; the included public projections preserve all measured cells. |
| Wide-query study | `ba5b4c8e` | Equal-work and live-width studies, 128 root/arm aggregates, phase sums and workload counts. |
| Matched-history width | `ab695a76` | All 128 warm/measured operations and process-level reconstruction for four fixed cells. |
| CPU consumer diagnostics | `af436df9` | Synthetic CPU width and frontier diagnostics, sources and public result summaries. These diagnose output/state distinctions, not GPU performance. |
| Implementation coverage | `10fc03ed` | Source-based account of online checks, capture observation, packed-node differences and failure boundaries. |
| Legacy-compatible RTX C | `ba019f30` | All timing samples for the first extracted candidate; complete-operator C/B is 1.1264. |

| Evidence | Fixed revision | Direct entry and role |
|---|---|---|
| Legacy-compatible RTX C1 | `304dcbff` | Separate-session candidate timings; complete-operator C1/B is 1.0261. |
| Old-contract U qualification | `01a4cf84` | Qualification against the older oracle fails; no timing under that contract. |
| Modern-contract U operator | `a6a92fb0` | Contract, wrapper, oracle, all 48 samples and analysis for the extracted modern alternative. |
| Default backend status | `d240c5b4` | Static/API result; complete backend numerical qualification and service timing are unrun. |
| Legacy numeric projections | `e8a4981e` | Earlier manuscript's `data`, `data_added` and figure provenance preserve structural, width, lifecycle and intervention inputs. Each copied file is separately hashed. |

### F.1 Public, restricted and unavailable material

Publicly included material comprises code for reconstruction and plotting, complete timing projections for the named panels, published root aggregates, selected diagnostic summaries, and their provenance. Original model files are not bundled; their identities and upstream licenses are recorded at the linked sources. Some original native logs and tensor data are company-restricted, and some temporary intermediates were pruned. A public summary is not a substitute for an unavailable full trace. No claim is made that all original native records can be downloaded or regenerated from this document alone.

The final archival research-data deposit, restriction wording and any authorized access mechanism require the author's confirmation. The package is sufficient for the stated CPU statistical and figure reconstruction. Running the original native experiments additionally requires the identified model bytes, hardware, numerical modules and authorized input access. The current preparation has not carried out those runs.

## References

[1] leecdiang. Implementation and companion evidence for fixed-reference parallel inference. GitHub repository, 2026. URL `https://github.com/leecdiang/HeteroSpec`. Experiment commits, evidence locations, and access scope are listed in the supplementary material.

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[3] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. arXiv:2103.03874v2, 2021. URL `https://arxiv.org/abs/2103.03874v2`.

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[6] llama.cpp contributors. Few-row quantized matrix multiplication in llama.cpp. Source code, ggml/src/ggml-cuda/mmq-few.cu, 2026. URL `https://github.com/ggml-org/llama.cpp/blob/0fb8f157702702f33f4e018de8701b2569a1bc28/ggml/src/ggml-cuda/mmq-few.cu`. Commit 0fb8f157702702f33f4e018de8701b2569a1bc28; accessed November 13, 2026.